

Lab3: Explore MVC web app in Asp.net

3.1 Create a template web app from VS 2022 using Model-View-Controller framework and add one page of your choice in the template app (e.g. About Us)

Explain the workflow of entire MVC app with special focus on Controller.

3.2 Setup for the backend connection (database with sqlite provide) in the temple app. Create Student model that defines student data type (at least three properties should be there) and dbcontext. Do migration to create necessary database and table.

Explain the EF core handling of the database interactions using ORM.

3.3 After successful step of 3.2, add StudentController in the app using scaffolding option provided by the VS to create entire CRUD operations.

Explain the StudentController's action methods to handle the respective views.

Lab4: Explore MVC web API in Asp.net

4.1 Create a template web app API from VS 2022.

Explain the REST web API and the workflow of Swagger api tester for Weatherforecast model.

4.2 Setup for the backend connection (database with sqlite provide) in API app created in 4.1. Create Student model that defines student data type (at least three properties should be there) and dbcontext. Do migration to create necessary database and table.

4.3 After successful step of 4.2, add StudentController API in the app using scaffolding option provided by the VS to create entire CRUD APIs.

Explain the StudentController's action methods for CRUD API tested by Swagger API tester.